

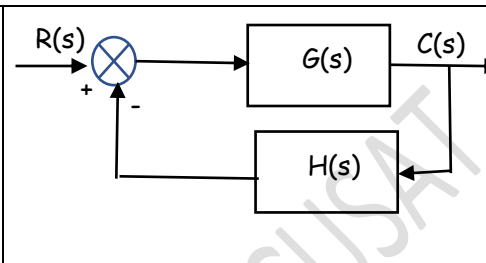
ROOT LOCUS METHOD

Root locus is a graphical representation of closed loop poles as a system parameter is varied. It is a powerful method of analysis of stability and transient response. We have learnt to design a first order or second order system for their transient responses by choosing appropriate gain and other system parameters. The real power of root locus technique lies in its ability to provide solutions for higher order systems.

Consider the system shown in figure. The closed loop transfer function

$$T(s) = \frac{C(s)}{R(s)} = \frac{G(s)}{1 + G(s)H(s)}$$

The characteristic equation of this system is $1 + G(s)H(s) = 0$ or $G(s)H(s) = -1 + j0$



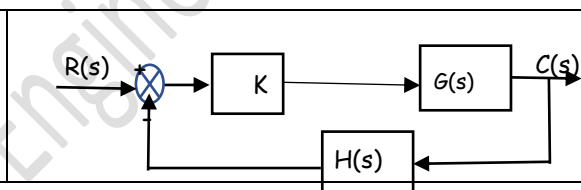
In most of the cases, $G(s)H(s)$ will be a polynomial in the complex variable 's'. Therefore, we can write the magnitude condition as $|G(s)H(s)| = 1$

and the angle condition as $\angle G(s)H(s) = \pm(2k+1)\pi$ (odd multiples of 180°) $K = 0, 1, 2, \dots$

The values of s that satisfies both the angle and magnitude conditions are the roots of the characteristic equation or the closed-loop poles.

In most cases, the forward path involves a gain parameter K so that the transfer function is

$$T(s) = \frac{C(s)}{R(s)} = \frac{KG(s)}{1 + KG(s)H(s)}$$



And the characteristic equation becomes $1 + KG(s)H(s) = 0$

The locus of the roots of this characteristic equation when the parameter K is varied from zero to infinity is known as the root locus.

Root Locus Procedure

Step 1: Write the characteristic equation in the form $1 + K \frac{\prod_{i=1}^m (s+z_i)}{\prod_{j=1}^n (s+p_j)} = 0 \gg K = -\frac{\prod_{j=1}^n (s+p_j)}{\prod_{i=1}^m (s+z_i)}$

For example consider $1 + K \frac{(s+1)}{s(s+2)(s+4)^2} = 0$

Step 2: Locate the poles of the closed loop transfer function

Root locus is the locus of the roots of the characteristic equation (poles) when the parameter K varies from 0 to ∞

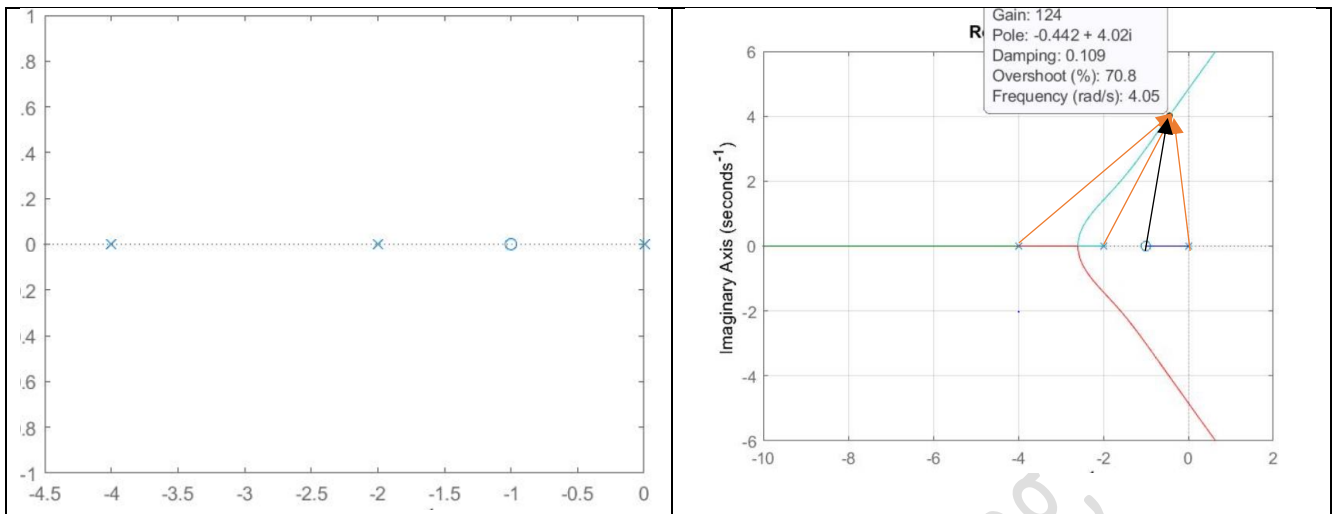
When $K = 0$, $-\frac{\prod_{j=1}^n (s+p_j)}{\prod_{i=1}^m (s+z_i)} = 0 \gg \prod_{j=1}^n (s+p_j) = 0$, the roots are simply the poles of $G(s)H(s)$.

When $K = \infty$, $-\frac{\prod_{j=1}^n (s+p_j)}{\prod_{i=1}^m (s+z_i)} = \infty \gg \prod_{i=1}^m (s+z_i) = 0$, the roots are simply the zeros of $G(s)H(s)$.

Therefore, the root locus starts from the poles and end at the zeros of $G(s)H(s)$ as K varies from zero to infinity. If the number of poles are n and number of zeros are m , usually n will be greater than m . Then $n - m$ zeros will lie at infinity.

The poles are $p_j = 0, -2, -4, -4$ and zeros are $z_i = -1$. Here we have four poles ($n = 4$) and one zero ($m = 1$). The root locus will start from the poles and end at the zeros. Three branches of the root loci will go to the zeros at infinity ($n - m = 3$)

Step 3: Locate the segments of the real axis that are part of the root loci. We know that any point on the root locus will satisfy the angle condition. The poles and root locus of the above example is shown in the figure below.



$$G(s)H(s) = \frac{(s+1)}{s(s+2)(s+4)^2}$$

$$\angle G(s)H(s) = \angle \frac{(s+1)}{s(s+2)(s+4)^2} = \angle (s+1) - \angle s - \angle (s+2) - \angle (s+4) - \angle (s+4)$$

Let us see if the point $s = -0.44 + j4.02$ is a point on the root locus. Substitute this value of s in the above expression.

$$\angle (s+1) = \angle (-0.44 + j4.02) + 1 = \angle 0.56 + j4.02$$

$$= \tan^{-1} \frac{4.02}{0.56} = 82.07^\circ$$

$$\angle (s) = \angle (-0.44 + j4.02) = \angle -0.44 + j4.02$$

$$= \tan^{-1} \frac{4.02}{-0.44} = -83.75^\circ \text{ (or } 96.25^\circ)$$

$$\angle (s+2) = \angle (-0.44 + j4.02) + 2 = \angle 1.56 + j4.02$$

$$= \tan^{-1} \frac{4.02}{1.56} = 68.79^\circ$$

$$\angle (s+4) = \angle (-0.44 + j4.02) + 4 = \angle 3.56 + j4.02$$

$$= \tan^{-1} \frac{4.02}{3.56} = 48.47^\circ$$

$$\angle G(s)H(s) = 82.07 - 96.25 - 68.79 - 48.47 - 48.47 = -180$$

Which is an odd multiple of 180. Hence the point $(-0.44 + j4.02)$ is on the root locus.

Now let us consider a point on the real axis (say $s = -0.5$)

$$\angle (s+1) = \angle (-0.5 + j0) + 1 = \angle 0.5 + j0 = \tan^{-1} \frac{0}{0.5} = 0^\circ$$

$$\angle (s) = \angle (-0.5 + j0) = \tan^{-1} \frac{0}{-0.5} = 180^\circ$$

$$\angle (s+2) = \angle (-0.5 + j0) + 2 = \angle 1.5 + j0 = \tan^{-1} \frac{0}{1.5} = 0^\circ$$

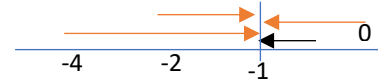
$$\angle (s+4) = \angle (-0.5 + j0) + 4 = \angle 3.5 + j0 = \tan^{-1} \frac{0}{3.5} = 0^\circ$$

$$\angle G(s)H(s) = 0 - 180 - 0 - 0 - 0 = -180$$

Which is an odd multiple of 180. Hence the point $(s = -0.5)$ is on the root locus. By this argument, any point between pole at 0 and zero at -1 will satisfy the angle criterion. Hence the real axis segment between $s=0$ and $s=-1$ will form part of the root loci.

Now let us consider another point on the real axis (say $s = -1.5$)

$$\begin{aligned}\angle(s+1) &= \angle(-1.5 + j0) + 1 = \angle -0.5 + j0 = \tan^{-1} \frac{0}{-0.5} = 180^\circ \\ \angle(s) &= \angle(-1.5 + j0) = \tan^{-1} \frac{0}{-0.5} = 180^\circ \\ \angle(s+2) &= \angle(-1.5 + j0) + 2 = \angle 0.5 + j0 = \tan^{-1} \frac{0}{0.5} = 0^\circ \\ \angle(s+4) &= \angle(-1.5 + j0) + 4 = \angle 2.5 + j0 = \tan^{-1} \frac{0}{2.5} = 0^\circ\end{aligned}$$

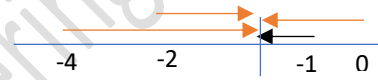


$$\angle G(s)H(s) = 180 - 180 - 0 - 0 - 0 = 0$$

Which is not an odd multiple of 180. Hence the point ($s = -1.5$) is not on the root locus. By this argument, any point between zero at -1 and pole at -2 will not satisfy the angle criterion. Hence the real axis segment between $s = -2$ and $s = -1$ will not form part of the root loci.

Consider another point on the real axis (say $s = -2.5$)

$$\begin{aligned}\angle(s+1) &= \angle(-2.5 + j0) + 1 = \angle -1.5 + j0 = \tan^{-1} \frac{0}{-1.5} = 180^\circ \\ \angle(s) &= \angle(-2.5 + j0) = \tan^{-1} \frac{0}{-2.5} = 180^\circ \\ \angle(s+2) &= \angle(-2.5 + j0) + 2 = \angle -0.5 + j0 = \tan^{-1} \frac{0}{-0.5} = 180^\circ \\ \angle(s+4) &= \angle(-2.5 + j0) + 4 = \angle 1.5 + j0 = \tan^{-1} \frac{0}{1.5} = 0^\circ\end{aligned}$$

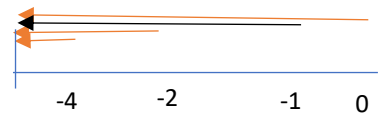


$$\angle G(s)H(s) = 180 - 180 - 180 - 0 - 0 = -180$$

Which is an odd multiple of 180. Hence the point ($s = -2.5$) is on the root locus. By this argument, any point between pole at -2 and poles at -4 will satisfy the angle criterion. Hence the real axis segment between $s = -4$ and $s = -2$ will form part of the root loci.

Consider another point on the real axis (say $s = -4.5$)

$$\begin{aligned}\angle(s+1) &= \angle(-4.5 + j0) + 1 = \angle -3.5 + j0 = \tan^{-1} \frac{0}{-3.5} = 180^\circ \\ \angle(s) &= \angle(-4.5 + j0) = \tan^{-1} \frac{0}{-4.5} = 180^\circ \\ \angle(s+2) &= \angle(-4.5 + j0) + 2 = \angle -2.5 + j0 = \tan^{-1} \frac{0}{-2.5} = 180^\circ \\ \angle(s+4) &= \angle(-4.5 + j0) + 4 = \angle -0.5 + j0 = \tan^{-1} \frac{0}{-0.5} = 180^\circ\end{aligned}$$



$$\angle G(s)H(s) = 180 - 180 - 180 - 180 - 180 = 3 \times (-180)$$

Which is an odd multiple of 180. Hence the point ($s = -4.5$) is on the root locus. By this argument, any point between pole at -4 and $s = -\infty$ will satisfy the angle criterion. Hence the real axis segment between $s = -4$ and $s = -\infty$ will form part of the root loci.

The real axis segment to the left of odd number of poles and/or zeros will be part of the root locus.

Step 4: Determine the separate loci (SL). Because the loci begin at the poles and end at the zeros, the number of separate loci is equal to the number of poles.

Step 5: Root loci must be symmetrical with respect to the real axis because the complex roots occur as conjugate pairs.

Step 6: The root loci proceed to the zeros at infinity along the asymptotes centered at σ_A and with angles ϕ_A . When the number of zeros m is less than the number of poles n , then $n - m$ sections of loci must end at infinity.

$$\sigma_A = \frac{\sum \text{poles} - \sum \text{zeros}}{n - m}$$

$$\phi_A = \frac{(2k + 1)180^\circ}{n - m}; \quad k = 0, 1, 2, \dots, (n - m - 1)$$

The centroid of the asymptotes can be found out from the characteristic equation $1 + KG(s)H(s) = 0$. $s = z_1 (s - z_1)(s - z_2)$

$$1 + K \frac{\prod_{i=1}^m (s + z_i)}{\prod_{j=1}^n (s + p_j)} = 0$$

$$1 + K \frac{[s^m - (z_1 + z_2 + \dots + z_m)s^{m-1} + \dots + (z_1 z_2 \dots z_m)]}{[s^n - (p_1 + p_2 + \dots + p_n)s^{n-1} + \dots + (p_1 p_2 \dots p_n)]} = 0$$

As $s \rightarrow \infty$; $1 + K \frac{[s^m - (\sum \text{zeros})s^{m-1}]}{[s^n - (\sum \text{Poles})s^{n-1}]} = 0$

$$1 + K \frac{s^{-m}[s^m - (\sum \text{zeros})s^{m-1}]}{s^{-m}[s^n - (\sum \text{Poles})s^{n-1}]} = 0$$

$$1 + K \frac{[1 - s^{-1}(\sum \text{zeros})]}{[s^{n-m} - s^{n-m-1}(\sum \text{Poles})]} = 0$$

The second term in the numerator is very small; we can use the binomial expansion (for $x < 1$)

$$1 - x = (1 + x)^{-1} = \frac{1}{1 + x}$$

$$1 + K \frac{1}{[s^{n-m} - s^{n-m-1}(\sum \text{Poles})][1 + s^{-1}(\sum \text{zeros})]} = 0$$

$$1 + K \frac{1}{[s^{n-m} + s^{n-m-1}(\sum \text{zeros}) - s^{n-m-1}(\sum \text{Poles}) - s^{n-m-2}(\sum \text{Poles})(\sum \text{zeros})]} = 0$$

$$1 + K \frac{1}{[s^{n-m} + s^{n-m-1}(\sum \text{zeros}) - (\sum \text{Poles})]} = 0$$

Again for large value of s , the characteristic equation $1 + K \frac{\prod_{i=1}^m (s + z_i)}{\prod_{j=1}^n (s + p_j)} = 0$ can be approximated as

$$1 + K \frac{(s - \sigma_A)^m}{(s - \sigma_A)^n} = 0 \gg 1 + K \frac{1}{(s - \sigma_A)^{n-m}} = 0 \gg 1 + K \frac{1}{s^{n-m} - \sigma_A(n-m)s^{n-m-1}} = 0$$

Comparing the coefficients in the two equations, we can write

$$-\sigma_A(n - m) = \sum \text{zeros} - \sum \text{Poles}$$

$$\sigma_A = \frac{\sum \text{Poles} - \sum \text{Zeros}}{n - m}$$

If a test point s is selected very far from the origin, $\angle G(s)H(s) = \pm(2k + 1)\pi$

$$\angle K \frac{(s - \sigma_A)^m}{(s - \sigma_A)^n} = \pm(2k + 1)\pi \quad \angle K \frac{1}{(s - \sigma_A)^{n-m}} = \pm(2k + 1)\pi \quad -(n - m) \angle (s - \sigma_A) = \pm(2k + 1)\pi$$

$$\angle (s - \sigma_A) = \pm \frac{(2k + 1)\pi}{n - m}$$

$$\sigma_A = \frac{\sum \text{poles} - \sum \text{zeros}}{n - m} = \frac{(-4 - 4 - 2 - 0) - (-1)}{4 - 1} = \frac{(-10) + 1}{3} = -3$$

$$\phi_A = \frac{(2k + 1)180^\circ}{n - m}; \quad k = 0, 1, 2$$

$$\phi_A = \frac{(2 \times 0 + 1)180^\circ}{3}, \frac{(2 \times 1 + 1)180^\circ}{3}, \frac{(2 \times 2 + 1)180^\circ}{3} = 60^\circ, 180^\circ, 300^\circ$$

Step 7: Determine the Break Away (Break in) Point. At the break away or break in points there will be multiplicity of roots. The value of s at the break away or break in point can be obtained as follows. Write the characteristic equation in the form $K = P(s)$. The roots of the polynomial $\frac{dK}{ds} = 0$ will give the break away or break in point.

The characteristic equation $1 + KG(s)H(s) = 0$ can be written as $1 + K \frac{Y(s)}{X(s)} = 0$

$$f(s) = X(s) + KY(s) = 0$$

This will have multiple roots at points where $\frac{df(s)}{ds} = 0$

$$X'(s) + KY'(s) = 0$$

$$K = -\frac{X'(s)}{Y'(s)}$$

Substituting this equation in $f(s)$, $X(s)Y'(s) - Y(s)X'(s) = 0$

When we solve this equation for s , we will get the location where multiple roots exists.

If we write $K \frac{Y(s)}{X(s)} = -1$ or $K = -\frac{X(s)}{Y(s)}$. If we find $\frac{dK}{ds}$ an equate to zero

$$\frac{dK}{ds} = -\frac{Y(s)X'(s) - Y'(s)X(s)}{Y^2(s)} = 0$$

We get the same equation $Y'(s)X(s) - Y(s)X'(s) = 0$

$$K = -\frac{s(s+2)(s+4)^2}{s+1} = -\frac{s^4 + 10s^3 + 32s^2 + 32s}{s+1}$$

$$\frac{dK}{ds} = -\frac{(s+1)(4s^3 + 30s^2 + 64s + 32) - (s^4 + 10s^3 + 32s^2 + 32s)}{(s+1)^2} = 0$$

$$3s^4 + 24s^3 + 62s^2 + 64s + 32 = 0$$

The roots are $-4, -2.5994, -0.7003 \pm j0.7317$

The break away point is -2.5994

Step 8: Determine the points where the root loci crosses the imaginary axis. This can be found out by use of the Routh's stability criterion

The characteristic equation of the given example is $1 + K \frac{(s+1)}{s(s+2)(s+4)^2} = 0$

$$s(s+2)(s+4)^2 + K(s+1) = 0$$

$$s^4 + 10s^3 + 32s^2 + (32+K)s + K = 0$$

s^4	1	32	K	
s^3	10	32+K		
s^2	$\frac{288-K}{10}$	K		
s^1	$\frac{K^2 - 156K - 9218}{28.8 - 0.1K}$			
s^0	K			

$$\frac{10 \times 32 - 1 \times (32 + K)}{10}$$

$$\frac{(28.8 - 0.1K) \times (32 + K) - 10K}{28.8 - 0.1K}$$

For $K=201.7$, we will have a row of zeros. Then the auxiliary polynomial will be $8.63s^2 + 201.7 = 0$

$$s = \pm j4.83$$

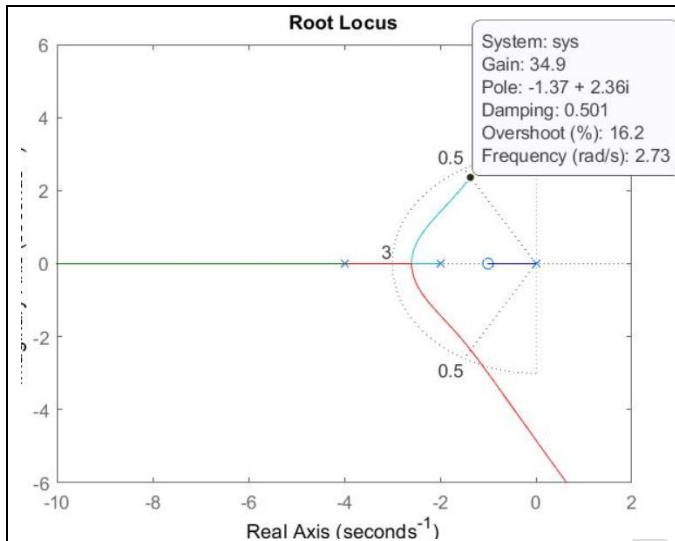
The root loci crosses the imaginary axis at $s = \pm j4.83$

Step 9: Determine the angle of departure from a complex pole and the angle of arrival of the locus at a complex zero, using the angle criterion.

Step 10: Choose test points in such a way that they satisfy the angle criterion. Draw the loci passing through these points.

Step 11: Determine the pair of dominant complex conjugate poles such that the damping ratio is 0.5.

Closed loop poles with $\zeta = 0.5$ lie on a line passing through the origin and making angles $\pm \cos^{-1} \zeta = \pm \cos^{-1}(0.5) = \pm 60^\circ$ with the negative real axis.



The closed loop poles are $-1.37 \pm j2.36$
The value of K that will yield these poles can be obtained from the magnitude condition

$$|KG(s)H(s)| = 1$$

$$K \left| \frac{(s+1)}{s(s+2)(s+4)^2} \right|_{s=-1.37+j2.36} = 1$$

$$K = \left| \frac{s(s+2)(s+4)^2}{s+1} \right|_{s=-1.37+j2.36}$$

$$K = \frac{|s||s+2||s+4||s+4|}{|s+1|} \quad s=-1.37+j2.36$$

$$K = \frac{|-1.37 + j2.36| |-1.37 + j2.36 + 2| |-1.37 + j2.36 + 4| |-1.37 + j2.36 + 4|}{|-1.37 + j2.36 + 1|}$$

$$K = \frac{\sqrt{(-1.37)^2 + (2.36)^2} \cdot \sqrt{(0.63)^2 + (2.36)^2} \cdot \sqrt{(2.63)^2 + (2.36)^2} \cdot \sqrt{(2.63)^2 + (2.36)^2}}{\sqrt{(-0.37)^2 + (2.36)^2}}$$

$$K = \frac{2.724 \times 2.442 \times 3.534 \times 3.534}{2.389} = 34.8$$

With this value of K, the characteristic equation becomes

$$s^4 + 10s^3 + 32s^2 + (32 + K)s + K = 0$$

$$s^4 + 10s^3 + 32s^2 + 66.8s + 34.8 = 0$$

The roots of this equation are $-6.5466, -0.7151, -1.37 + j2.36, -1.37 - j2.36$

***Example**

Sketch the root locus of the system if $G(s)H(s) = \frac{K}{s(s^3+12s^2+64s+128)}$

Step 1:

Write the characteristic equation in the form $1 + G(s)H(s) = 0$; $1 + \frac{K}{s(s^3+12s^2+64s+128)} = 0$

Step 2:

Determine the poles and zeros of $G(s)H(s)$. The poles are $s = 0, -4, -4 \pm j4$. There are no zeros. The number of poles (n) = 4.

The number of zeros (m) = 0.

The root loci originate from the poles of $G(s)H(s)$ and end at the zeros.

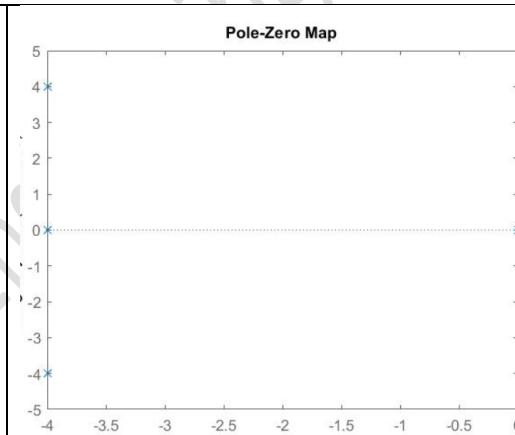
The number of branches of root loci proceeding to the zeros at infinity = $n - m = 4$

Step 3:

Number of separate loci is equal to the number of poles (n)

Step 4:

Locate the segments of real axis that are part of the root loci. The real axis segment to the left of odd number of poles and zeros will be part of the root loci. The first pole/zero is located at the origin. So the part of real axis from $s = 0$ till the next pole/zero at $s = -4$ will be part of the loci. Since there are no more poles/zeros in the real axis after $s = -4$, no more part of real axis will form a part of root loci.



Step 5:

The root loci are symmetrical about the real axis

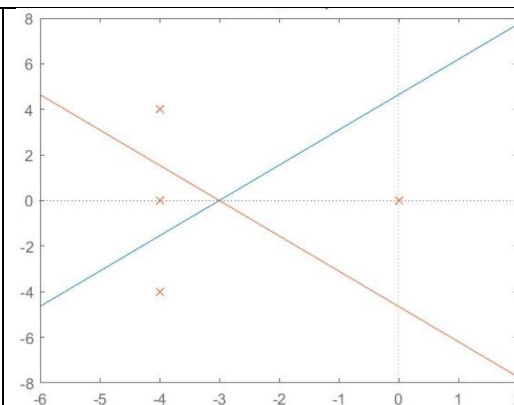
Step 6:

The root loci proceed to the zeros at infinity along the asymptotes centered at

$$\sigma_A = \frac{\sum \text{poles} - \sum \text{zeros}}{n-m} = \frac{(0-4-4+j4-j4)-(0)}{4-0} = \frac{-12}{4} = -3$$

$$\phi_A = \frac{(2k+1)180^\circ}{n-m}; k = 0, 1, 2, 3$$

$$\phi_A = \frac{180^\circ}{4}, \frac{3 \times 180^\circ}{4}, \frac{5 \times 180^\circ}{4}, \frac{7 \times 180^\circ}{4} = 45^\circ, 135^\circ, 225^\circ, 315^\circ$$



Step 7:

Determine the break away/break in point.

Write the characteristic equation $K = P(s)$

Determine the roots of $\frac{dK}{ds} = 0$

$$1 + \frac{K}{s(s^3+12s^2+64s+128)} = 0$$

$$K = -s(s^3 + 12s^2 + 64s + 128)$$

$$\frac{dK}{ds} = -(4s^3 + 36s^2 + 128s + 128) = 0$$

$$s = -3.7117 \pm j2.5533, -1.5767$$

We take the value $s = -1.57$ for the breakaway point

Step 8:

Determine the imaginary axis crossing by constructing the Routh table.

The characteristic equation is $1 + \frac{K}{s(s^3+12s^2+64s+128)} = 0$

$$s^4 + 12s^3 + 64s^2 + 128s + K = 0$$

s^4	1	64	K
s^3	12	128	
s^2	53.33	K	
s^1	$\frac{53.33 \times 128 - 12 \times K}{53.33}$		
s^0	K		

$$\frac{12 \times 64 - 1 \times 128}{12} = 53.33$$

$$\frac{12 \times K - 1 \times 0}{12} = K$$

$$\frac{53.33 \times 128 - 12 \times K}{53.33}$$

When the root loci cross the imaginary axis, there will be a row of zeros. This is possible only in the s^1 row when $53.33 \times 128 - 12 \times K = 0$ or $K = 568.89$.

The auxiliary polynomial is $53.33s^2 + 568.89 = 0$

Or $s = \pm j3.26$. is the point where the root loci crosses the imaginary axis

Step 9:

Determination of the angle of locus departure from the complex poles and angle of locus arrival at complex zeros from the angle criterion.

Consider a test point s very close to the complex pole at $-4 + j4$. Let θ_1 be the angle subtended by the vector from the pole at $-4 + j4$ to the test point s

θ_2 from -4 to $s = 90^\circ$

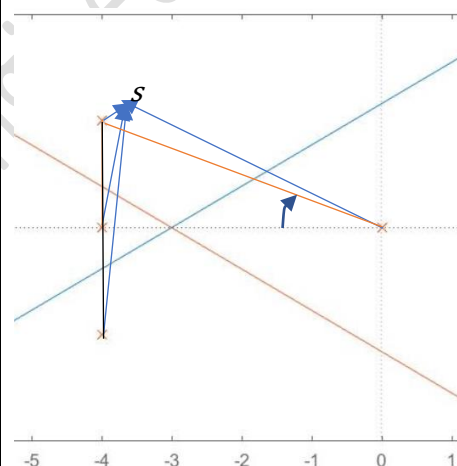
θ_3 from $-4 - j4$ to $s = 90^\circ$

θ_4 from 0 to $s = 135^\circ$ ($\tan^{-1} \frac{4}{4} = 45^\circ$)

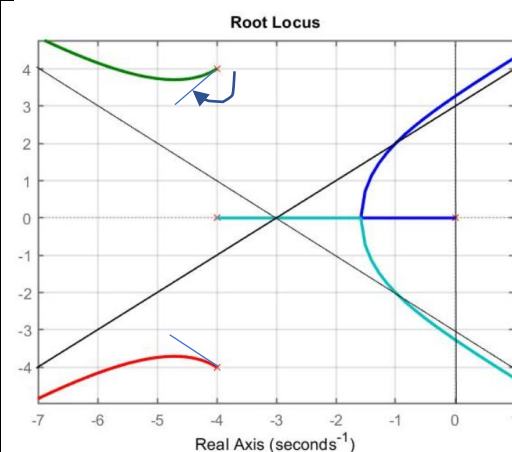
$\theta_1 + \theta_2 + \theta_3 + \theta_4 = \text{odd multiple of } 180^\circ$

$$\theta_1 = 180 - 90 - 90 - 135 = -135^\circ$$

Since the root loci is symmetrical about the real axis, the test point s in the vicinity of the pole at $-4 - j4$ will have an angle contribution of $+135^\circ$

**Step 10:**

Determine the root locations which satisfy the angle criterion and join such points to get the locus.



#Exercise 36

Sketch the root locus of the system if $G(s)H(s) = \frac{K}{s(s+4)(s^2+2s+2)}$. Verify your answer by plotting it in Matlab/Scilab.

#Exercise 37

Sketch the root locus of the system if $G(s)H(s) = \frac{K(s+6)}{s(s+3)(s^2+2s+2)}$. Verify your answer by plotting it in Matlab/Scilab.

#Exercise 38

Sketch the root locus of the system if $G(s)H(s) = \frac{K(s+4)(s+6)}{s(s+2)}$. Verify your answer by plotting it in Matlab/Scilab.

#Exercise 39

Sketch the root locus of the system if $G(s)H(s) = \frac{K}{s(s+3)(s+5)(s^2+2s+2)}$. Verify your answer by plotting it in Matlab/Scilab.

#Exercise 40

Using Matlab/Scilab, sketch the root locus of the system whose $G(s)H(s) = \frac{K}{s(s+5)}$. Observe the changes in the root locus when it is modified by adding poles and zeros as given below

- i) $G(s)H(s) = \frac{K}{s(s+5)(s+10)}$
- ii) $G(s)H(s) = \frac{K(s+1)}{s(s+5)}$
- iii) $G(s)H(s) = \frac{K(s+10)}{s(s+5)}$
- iv) $G(s)H(s) = \frac{K(s+1)}{s(s+5)(s+10)}$
- v) $G(s)H(s) = \frac{K(s+1)(s+10)}{s(s+5)(s+3)}$

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